Blueprint

Huddling, diatonic and pentatonic scales in Z12: Fourier maxima and convex-energy minima formalized in Lean 4

Mathematical music theory · discrete Fourier analysis on $\mathbb{Z}_{12}$

Carles Marín

Statements

Theorem 1 — Huddling in $\mathbb{Z}_{12}$ 4
Theorem 2 — Diatonic energy minimum 4
Theorem 3 — Fourier–energy equivalence 5
Theorem 4 — Pentatonic extremality 5

At a glance

What the paper contributes

- A connected Lean 4 formalization, with complete twelve-tone equality conditions, of why the seven white keys solve two optimization problems: they maximize $|\widehat{A}(5)|$ and minimize every strictly admissible convex pair energy (Theorem 3).
- The route: Huddling in $\mathbb{Z}_{12}$ (Theorem 1), transported to frequency five by M_5; double Abel summation for the energy (Theorem 2); complementation carries both to the pentatonic sets (Theorem 4).
- Exact arithmetic throughout: comparisons in $\mathbb{Z}[\sqrt{3}]$, no rounded trigonometric values. Two finite censuses use `native_decide` (trusted compiled evaluation), stated in the axiom audit.
- The mathematics is classical; the paper claims the connected formalization, not a new characterization.

🚀 **Start here: Theorem 1**: consecutive arcs maximize the first Fourier magnitude, exactly on their transposition orbit.

Main results:

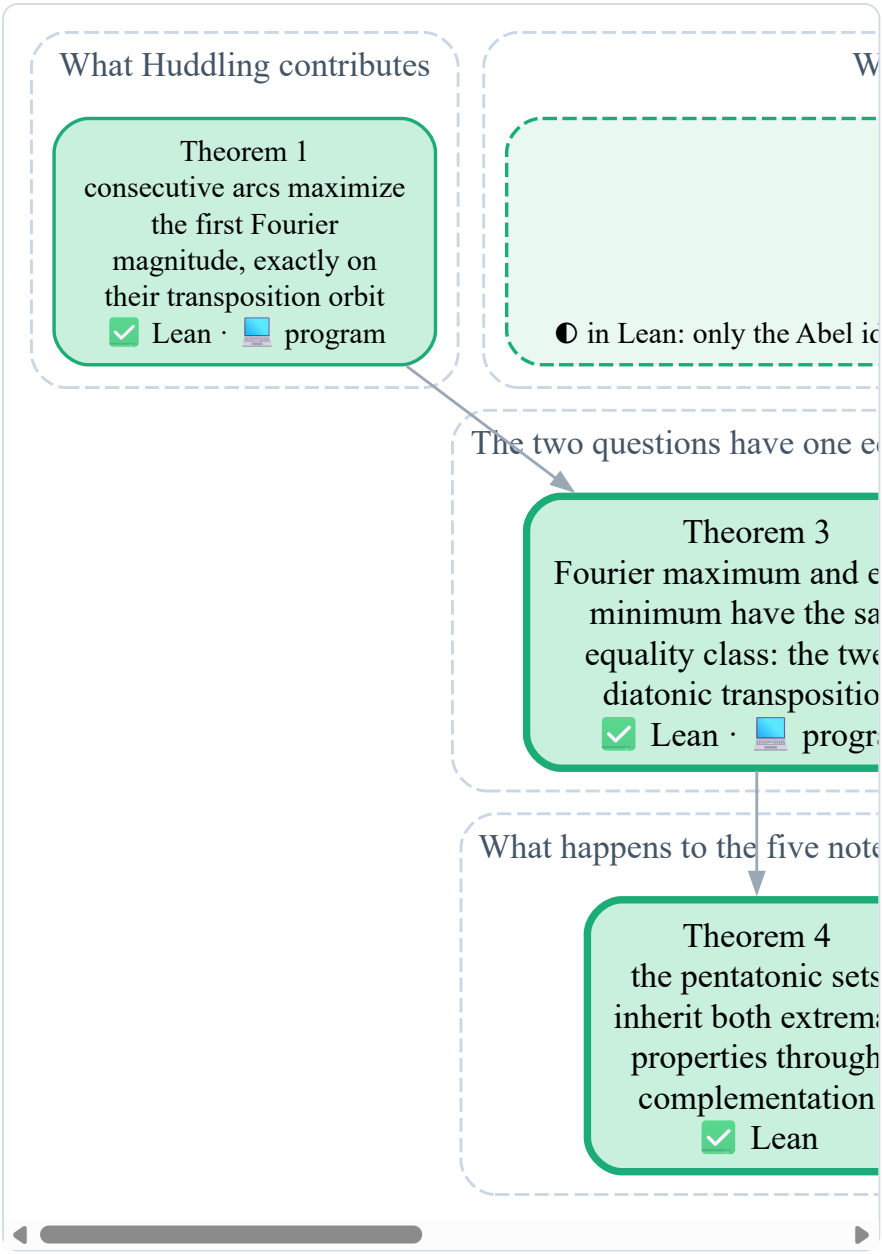
- **Theorem 3**: Fourier maximum and energy minimum have the same equality class: the twelve diatonic transpositions
- **Theorem 4**: the pentatonic sets inherit both extremal properties through complementation

4 results | ✅ **3** formalized in Lean | ◐ **1** with a part in Lean

💻 **2** checked by a program | written proof only (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



📐 Setting (copied from the paper, §2)

For an unweighted set $A \subseteq \mathbb{Z}_{12}$, use the unnormalized DFT

$$\widehat{A}(t) = \sum_{a \in A} e^{-2\pi i a t/12}, \qquad S_t(A) = |\widehat{A}(t)|^2.$$

Transposition $T_s A = A + s$ changes phase and preserves S_t. For a unit $u \in \mathbb{Z}_{12}^{\times}$, put $M_u A = \{ua : a \in A\}$. Reindexing the finite sum gives $\widehat{M_u A}(t) = \widehat{A}(ut)$. For $C_7 = \{0,1,2,3,4,5,6\}$ and $D = \{0,2,4,5,7,9,11\}$, $T_{11} M_5 C_7 = D$ and $M_5 D = T_7 C_7$.
(§4) Write $d(a,b) = \min((a-b) \bmod 12, (b-a) \bmod 12)$. For a real potential $V : \{1,\dots,6\} \to \mathbb{R}$, the all-pairs energy is $E_V(A) = \sum_{\{a,b\} \subseteq A} V(d(a,b)) = \sum_{j=1}^{6} i_j(A) V(j)$. We call V admissible when $\Delta^2 V(j) := V(j) - 2V(j+1) + V(j+2) \geq 0$ $(1 \leq j \leq 4)$ and $V(6) \leq V(5)$. Strict admissibility means that all five displayed inequalities are strict.

3 What Huddling contributes

Theorem 1. Lean **(Huddling in $\mathbb{Z}_{12}$)**✓ p. 2

> *For every $A \subseteq \mathbb{Z}_{12}$, with $k = |A|$ and $C_k = \{0, \dots, k-1\}$,*
>
> $$S_1(A) \le S_1(C_k), \qquad S_1(A) = S_1(C_k) \iff A = T_s C_k \text{ for some } s \in \mathbb{Z}_{12}.$$
>
> *The same inequality and equality condition hold for Fourier magnitudes.*

✅ Checked when this page was built: Huddling.lean, MTransform.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — arc_maximizes_power, arc_unique, powerSpec_one_re_le_iff: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean Huddling.lean MTransform.lean` (SHA-256 of the compiled file: `e53d6d27167c0ac7`…; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** classical: the roots-of-unity lemma recorded by Amiot [Amiot2009, Lemma 2]; the paper certifies its twelve-tone instance with the complete equality class.

↳ Used in: Theorem 3 (p. 5)

Proof

> Formalized in Lean (see the linked declarations).

🛡️ **Evidence & scope**

- **Program:** all 4096 subsets, by integer pairs (P,Q) in $\mathbb{Z}[\sqrt{3}]$. `huddling_verify.py`
- **Program:** all 4096 subsets, Fourier sums in $\mathbb{Q}(\zeta_{12})$ with exact algebraic-real comparison (SageMath). `sage/huddling_check.py`

4 Why the energy selects the same collection

Theorem 2. (Diatonic energy minimum) p. 4

> *For every admissible V and every $A \subseteq \mathbb{Z}_{12}$ with $|A| = 7$, $E_V(D) \le E_V(A)$. If V is strictly admissible, then*
>
> $$E_V(A) = E_V(D) \iff A \in \mathcal{O}_D.$$

✅ Checked when this page was built: AllPairsEvenness.lean, DiatonicExtremality.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — abel_bridge, diatonic_energy_unique: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean AllPairsEvenness.lean DiatonicExtremality.lean` (SHA-256 of the compiled file: `b6d144324b2b8efa`…; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** classical viewpoint (Douthett–Krantz [DK]); the contribution is the formalization with the complete equality class.

↳ Used in: Theorem 3 (p. 5)

5 The two questions have one equality class

Theorem 3. Lean **(Fourier–energy equivalence)**✓ p. 5

> *For every strictly admissible V and every seven-element $A \subseteq \mathbb{Z}_{12}$,*
>
> $$\boxed{|\widehat{A}(5)| = 2 + \sqrt{3} \quad \Longleftrightarrow \quad E_V(A) = E_V(D).}$$
>
> *Both sides hold precisely for the twelve transpositions of D.*

✅ Checked when this page was built: DiatonicExtremality.lean, Huddling.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — spectral_energy_equivalence, diatonic_norm_unique, fifth_frequency_unique: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean DiatonicExtremality.lean Huddling.lean` (SHA-256 of the compiled file: `b39e90bd655cefdb…`; a source page on this site may show it with comments edited, under its own hash).

> 🆕 **Classical and new:** the paper calls it a corollary of published equality-class characterizations, not a new characterization (§9).

↳ Used in: Theorem 4 (p. 5)

Proof

> Equation (4) identifies the left-hand equality class with $\mathcal{O}_D$. Theorem 2 identifies the right-hand equality class with $\mathcal{O}_D$. This is `DiatonicExtremality.spectral_energy_equivalence`.

🛡️ **Evidence & scope**

- **Program:** all 792 seven-note sets at frequency five, two arithmetic routes. `huddling_verify.py`, `sage/huddling_check.py`

6 What happens to the five notes left over?

Theorem 4. Lean **(Pentatonic extremality)**✓ p. 6

> *For every five-element $A \subseteq \mathbb{Z}_{12}$, $|\widehat{A}(5)| \leq 2 + \sqrt{3}$. For every admissible V, $E_V(P) \leq E_V(A)$. Under strict admissibility,*
>
> $$|\widehat{A}(5)| = 2 + \sqrt{3} \quad \Longleftrightarrow \quad E_V(A) = E_V(P) \quad \Longleftrightarrow \quad A = T_s P \text{ for some } s.$$

✅ Checked when this page was built: ExtremalityConnections.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — pentatonic_spectral_energy_equivalence, pentatonic_energy_gap, energy_compl: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean`

`ExtremalityConnections.lean` (SHA-256 of the compiled file: `1c529e4f4155bab6…`; a source page on this site may show it with comments edited, under its own hash).

> 🆕 **Classical and new:** complement-energy transport is covered in greater generality by Bushaw, Cody and Leffler [BCL, Thms 2.16 and 3.5].

↳ Not used by another result of the paper.

Proof

> Formalized in Lean (see the linked declarations).

References

1

D. Lewin, “Re: Intervallic relations between two collections of notes,” *Journal of Music Theory* **3**(2) (1959), 298–301. doi:10.2307/842856.

2

J. Clough and J. Douthett, “Maximally even sets,” *Journal of Music Theory* **35**(1/2) (1991), 93–173. doi:10.2307/843811.

3

E. Amiot, “David Lewin and maximally even sets,” *Journal of Mathematics and Music* **1**(3) (2007), 157–172. doi:10.1080/17459730701654990.

4

E. Amiot, “Gammes bien réparties et transformée de Fourier discrète,” arXiv:math/0604210 (2006), Theorems 2, 6, and 7. https://arxiv.org/abs/math/0604210.

5

E. Amiot, “Discrete Fourier Transform and Bach’s Good Temperament,” *Music Theory Online* **15**(2) (2009). https://mtosmt.org/issues/mto.09.15.2/mto.09.15.2.amiot.html.

6

J. Douthett and R. Krantz, “Maximally even sets and configurations: common threads in mathematics, physics, and music,” *Journal of Combinatorial Optimization* **14** (2007), 385–410. doi:10.1007/s10878-006-9041-5.

7

N. Bushaw, B. Cody, and C. Leffler, “Sets of vertices with extremal energy,” arXiv:2407.18785 (2024), version 3 (2025), Theorems 2.16 and 3.5. https://arxiv.org/abs/2407.18785.

8

E. Amiot and W. A. Sethares, “An algebra for periodic rhythms and scales,” author manuscript (2009), Section 4.2.2. https://sethares.engr.wisc.edu/paperspdf/AlgofScales.pdf.

9

C. Marín Muñoz, “The per-voice Fourier signature: the bass is the spectrally purest voice,” music-math series (2026). doi:10.5281/zenodo.20971090.

10

C. Marín, “The phase taxonomy of pitch-class set invariants,” music-math series, Note 2 (2026). doi:10.5281/zenodo.20826774.

11

C. Marín, “A tritone-centered self-duality: the set class 6-30 and its order-12 neo-Riemannian dual,” music-math series, Note 1 (2026). doi:10.5281/zenodo.20820962.

12

C. Marín, “The Tonnetz spectrum is the generating triad’s Fourier balance profile,” music-math series, Note 3 (2026). doi:10.5281/zenodo.20862822.

13

The Mathlib Community, *Mathlib4*, with the `ZMod.dft` development by D. Loeffler. https://github.com/leanprover-community/mathlib4.

14

L. Aniva and C. Wang, *Prismriver: Formalization of Music Theory and Algorithmic Composition in Lean 4*, arXiv:2606.19936 (2026). https://github.com/lenianiva/Prismriver.